TOPIC 2 Systems of Linear Equations and Inequalities

Name _____ Date _____

Reminders

System of inequalities: graph and shade each one; the overlap is the solution set.

A solution makes EVERY inequality true. Points on a dashed line are not solutions.

In a story: no negative amounts, so stay in the first quadrant.

1. Where is the solution set of $y > \frac{2}{3}x + 1$ and $y < -\frac{2}{3}x + 1$?

Tip: Sketch both lines, then shade each.

A. between the lines, left of where they cross

B. between the lines, right of where they cross

C. above both lines

2. Which point is a solution of $y > \frac{2}{3}x + 1$ and $y < -\frac{2}{3}x + 1$?

Tip: Test each point in both inequalities.

F. (3, 0)

G. (−3, 0)

H. (0, 1)

3. Explain why (0, 1) is not a solution of $y > \frac{2}{3}x + 1$ and $y < -\frac{2}{3}x + 1$.

> **Tip:** What does a dashed line mean?

Test y > ⅔x + 1: 1 > _____ is _____

So (0, 1) is _____ a solution